

100+ Linux Technical Interview Questions:

1. **Give any 3 examples of operating systems.**
 - Windows, Linux, and macOS.
2. **What is the root home directory?**
 - The root home directory is /root.
3. **Your company has terminated a server administrator. What is the first thing as an administrator you should do to enhance the security?**
 - Change the root password.
4. **How to check kernel version?**
 - Use the command `uname -a`.
5. **How to become a root user from a regular user?**
 - Use `su -` or `sudo su` to switch to the root user.
6. **How to check the computer name or hostname in Linux?**
 - Use the `hostname` command.
7. **List 3 basic commands to navigate the filesystem.**
 - `cd`, `pwd`, and `ls`.
8. **How to check network interfaces in Linux?**
 - Use `ifconfig`, `ip addr`, or `ip a`.
9. **List 3 different methods of adding a user.**
 - Methods include using `useradd`, `adduser`, or manually editing `/etc/passwd` and `/etc/shadow` files.
10. **What is the command to create a new user?**
 - `useradd username`.
11. **How to change a user password?**
 - Use the command `passwd username`.
12. **Which directory has all the configuration files?**
 - The `/etc` directory.
13. **List any 4 Linux distributions.**
 - Kali, Red Hat, CentOS, Ubuntu, and SUSE.
14. **How to log off from the system?**
 - Use the `exit` command.
15. **How to create a directory?**
 - Use the `mkdir` command.
16. **Explain the purpose of the “touch” command.**
 - `touch` is used to create an empty file or update the timestamp of an existing file.
17. **If a command hangs, how to stop it and get the prompt back?**
 - Press `Ctrl + C`.
18. **Which command is used to count words or lines?**
 - `wc`.
19. **How to rename a file or directory?**
 - Use the `mv` command.

20. How to change a hostname in Linux?

- Edit /etc/hostname and /etc/hosts, then restart the system or use hostnamectl set-hostname newname.

21. What is the command to change file/directory permissions?

- Use chmod.

22. What is the purpose of pipe (|)?

- To pass the output of one command as input to another command.

23. What is /etc directory used for?

- It contains configuration files for the system.

24. Which command is used to list files in a directory?

- ls -l.

25. There is a command which gives you information about other commands, please explain that command and what is it used for?

- man is used to display the manual pages for other commands, providing detailed information on usage and options.

26. How to delete a file and a directory?

- Use rm filename for files and rmdir dirname for directories.

27. What is the difference between “tail” and “tail -10”?

- tail displays the last 10 lines of a file by default, while tail -10 explicitly specifies to display the last 10 lines.

28. List 4 commands to display or read a file's contents.

- cat, more, less, vi.

29. Which command is used to read the top 5 lines of a file?

- head -5 filename.

30. What are the different commands or methods to write to a file?

- echo > filename, vi filename, cat > filename.

31. What are the different types of shells?

- sh, bash, ksh, csh, zsh.

32. Which is the core of the operating system?

- Kernel.

33. Which among the following interacts directly with system hardware?

- Kernel.

34. List a few commands that are used in troubleshooting network-related issues?

- netstat: Displays network connections.
- tcpdump: Captures and analyzes network traffic.
- ping: Tests connectivity to another host.
- traceroute: Traces the route packets take to a network host.

35. How to combine 2 files into 1? E.g., you have 3 lines in file “A” and 5 lines in file “B”, which command syntax to use that will combine into one file of 3+5 = 8 lines?

- cat fileA fileB > combinedfile.

36. What is the echo command used for?

- The echo command is used to display a line of text or output a string to the screen.

37. What does the following command do?

- `echo "This year the summer will be great" > file1`
- It creates a new file named "file1" and writes the text "This year the summer will be great" into it. If the file already exists, it will be overwritten.

38. When you login you get "\$" prompt, what is the prompt for root?

- The prompt for the root user is typically #.

39. Explain the difference between grep and egrep.

- `grep` is used for basic regular expression pattern matching, while `egrep` (or `grep -E`) supports extended regular expressions, which allow more complex pattern matching.

40. What is the port number for DNS, NTP, and NFS?

- DNS: 53, NTP: 123, NFS: 111 (portmapper) and 2049 (NFS).

41. What is the configuration file name of DNS and where is it located?

- The DNS configuration file is named `named.conf` and is typically located in `/etc`.

42. How many new directories will be created after running the following command `mkdir {a..c}{1..3}`?

- 9 directories will be created: `a1`, `a2`, `a3`, `b1`, `b2`, `b3`, `c1`, `c2`, `c3`.

43. Your PC is configured with a DNS server address but not the default gateway. Can the PC access the internet?

- No, without a default gateway, the PC cannot access the internet.

44. What is the difference between IP and Gateway?

- An IP address is a unique identifier for a device on a network, while a gateway is a network point that acts as an entrance to another network, often used to connect local networks to external networks like the internet.

45. Can you assign one static IP to 2 computers, if not then why?

- No, assigning the same static IP to two computers will create an IP conflict, causing network communication issues.

46. How to change IP address to static?

- You can change the IP address to static by editing the network configuration files, such as `/etc/network/interfaces` on Debian-based systems or `/etc/sysconfig/network-scripts/ifcfg-eth0` on Red Hat-based systems, and setting the IP address manually.

47. You are trying to ping a server by hostname and you get an error message, "ping: unknown host ...". What could be the reason and how to solve the problem so you can ping it by hostname?

- The issue could be due to a missing hostname-to-IP mapping in the `/etc/hosts` file or an incorrect DNS configuration. Check and update these files to resolve the issue.

48. Explain the difference between relative and absolute path.

- An absolute path starts from the root directory `/` and specifies the complete path to a file or directory. A relative path is based on the current directory and does not start with `/`.

49. What is the command to change file/directory ownership and group?

- Use `chown` to change ownership and `chgrp` to change the group.
50. **List any 3 types of filesystem.**
- `ext4`, `NTFS`, and `FAT`.
51. **When you login you get a message on the screen. What is the name of that file and where is it located?**
- The file is `/etc/motd` (Message of the Day).
52. **What is the `/bin` directory used for?**
- The `/bin` directory contains essential binary executables needed for booting and system repair.
53. **What are the different types of DNS Server?**
- Master (Primary) and Secondary (Slave) DNS servers.
54. **Where are the zone files located for DNS service?**
- Typically located in `/var/named/zonefiles`.
55. **How many megabytes in 1 gigabyte?**
- There are 1024 megabytes in 1 gigabyte.
56. **What is the purpose of having different network ports?**
- Different network ports allow multiple applications to communicate over the network simultaneously without interference.
57. **How to display the first column of a file?**
- Use `cat filename | awk '{print $1}'` or `awk '{print $1}' filename`.
58. **What is the name of the DNS rpm package?**
- The package is called `bind`.
59. **What is the difference between `nslookup` and `dig` commands?**
- `nslookup` is a simple tool for querying DNS servers, while `dig` provides more detailed information and is more flexible for DNS troubleshooting.
60. **How to check your user id and group id?**
- Use the `id` command.
61. **What is the difference between “`kill`” and “`kill -9`” command?**
- `kill` sends a signal to terminate a process gracefully, allowing it to clean up resources. `kill -9` sends a `SIGKILL` signal, forcing the process to terminate immediately without cleanup.
62. **What is a subnet?**
- A subnet is a segmented piece of a larger network, designed to improve performance and security by grouping devices with similar network requirements.
63. **You are troubleshooting an issue with Redhat support and they have asked you to send the contents of `/etc` directory. How and which method will you use to transfer the contents?**
- Compress the `/etc` directory using `tar` and transfer it using `ftp` or `scp`.
64. **What is `syslogd` daemon and its purpose?**
- `rsyslogd` is a system utility providing support for message logging. It is an enhanced version of `syslogd`.
65. **What is the command to untar a tarred file?**

- Use `tar -xvf filename.tar`.

66. What is the /proc directory used for?

- The /proc directory contains virtual files that provide a view into the kernel's view of the system.

67. What is the purpose of the nsswitch.conf file?

- It specifies the sources from which to obtain name-service information in a range of categories and in what order.

68. Which service/daemon should be running on the server that allows you to connect remotely?

- sshd (SSH Daemon).

69. What is the purpose of a firewall?

- A firewall controls incoming and outgoing network traffic based on predetermined security rules.

70. List any 3 IT components.

- Hardware, Operating System, and Applications.

71. Which directory has all the commands we use, e.g., ls, cd, etc.?

- /usr/bin or /bin.

72. What is the difference between memory, virtual memory, and cache?

- Memory (RAM) is the physical hardware inside a computer that temporarily stores data.
- Virtual memory is a memory management capability that uses disk space as an extension of RAM.
- Cache is a smaller, faster memory component that stores copies of frequently accessed data for quick access.

73. Correct order of interaction:

- a. User >> Operating System >> Hardware.

74. Which of the following is a communication command?

- mail.

75. Why is the “tail -f logfilename” command used most often and what does it do?

- It outputs all incoming logs in real-time, useful for monitoring log files as they are updated.

76. How to sort a file in reverse order?

- Use `sort -r filename`.

77. List all byte sizes from smallest to largest.

- Bit, Byte, Kilobyte (KB), Megabyte (MB), Gigabyte (GB), Terabyte (TB), Petabyte (PB), Exabyte (EB).

78. How to check the total number of partitions in Linux?

- Use `fdisk -l`.

79. How to access a Linux system from a Linux system?

- Use `ssh`.

80. Explain the procedure of bonding 2 NICs or interfaces together.

- Use network bonding to combine two or more NICs into a single bonded interface for redundancy or increased throughput. This typically involves

configuring /etc/network/interfaces or /etc/sysconfig/network-scripts/ifcfg-bond0 and using the bonding kernel module.

81. **What is the exact command syntax to list the 5th column of a file and cut the first 3 letters?**
 - `awk '{print $5}' filename | cut -c1-3.`
82. **What is the /etc/hosts file used for?**
 - It is used to resolve hostnames to IP addresses locally.
83. **List any 3 options of 'df' command and what they are used for.**
 - -h: human-readable format, -i: inodes information, -T: file system type.
84. **What is swap space and how to check swap space?**
 - Swap space is a portion of the hard drive used as an extension of RAM. Check swap space using `swapon -s` or `free`.
85. **What is inode and how to find an inode of a file?**
 - An inode is a data structure on a filesystem that stores information about a file or directory. Use `ls -li filename` to find the inode number.
86. **Which file to edit for kernel tuning?**
 - Edit /etc/sysctl.conf for kernel parameter tuning.
87. **What is the latest version of Redhat?**
 - Search online for the most recent version as it frequently updates.
88. **Name the command to find a specific word from a file.**
 - `grep word filename.`
89. **You have scheduled a job using crontab but it does not run at the time you specified, what could be the reason and how would you troubleshoot?**
 - Check system time, crontab entry syntax, and /var/log/cron for errors.
90. **How to check system hardware information?**
 - Use `dmidecode`.
91. **How to check network interface MAC address?**
 - Use `ifconfig` or `ip link`.
92. **If I don't want others to read my file1, how do I do that?**
 - Remove read permission for others using `chmod o-r file1`.
93. **What is the purpose of "uniq" and "sed" commands?**
 - `uniq` removes duplicate lines from sorted input, `sed` is a stream editor for filtering and transforming text.
94. **Which command is used to list the contents of a directory in the most recent time and in reverse order, meaning the most updated file should be listed on the bottom?**
 - `ls -ltr`.
95. **What is the difference between tar, gzip, and gunzip?**
 - `tar` is used for archiving files, `gzip` compresses files, and `gunzip` decompresses files.
96. **What are the different ways to install an OS?**
 - Using a DVD, DVD ISO, or network boot.
97. **How to view the difference between two files?**

- Use diff file1 file2.

98. You noticed that one of the Linux servers has no disk space left, how would you troubleshoot that issue?

- If using LVM, add more disk space and extend the logical volume. If not, add a new disk, create a partition, and link it to an existing filesystem.

99. How to check Redhat version release?

- Use cat /etc/redhat-release or uname -a.

100. What is the difference between TCP and UDP?

- TCP is connection-oriented and reliable, while UDP is connectionless and faster but less reliable.

101. What is a zombie process?

- A zombie process is a process that has completed execution but still has an entry in the process table, waiting for the parent process to read its exit status.

102. How do you search for a pattern/word in a file and then replace it in an entire file?

- Use sed for search and replace operations.

103. How to check the number of users logged in?

- Use who command.

104. What is the command to view the calendar of 2024?

- cal 2024.

105. Which command is used to view disk space?

- df -h.

106. How to create a new group in Linux?

- groupadd groupname.

107. What is the command to send a message to everyone who is logged into the system?

- Use wall command.

108. Which command is used to check the total number of disks?

- fdisk -l.

109. What is a mail server record in DNS?

- MX (Mail Exchange) record.

110. What does the following command line do? ps -ef | awk '{print \$1}' | sort | uniq

- Lists the first column of all running processes, sorts them, and removes duplicates.

111. You get a call that when a user goes to www.yourwebsite.com it fails and gets an error, how do you troubleshoot?

- Check user internet connectivity, DNS configuration, server status, and web service availability.

112. List 4 different directories in /?

- /etc, /bin, /tmp, /home.

